Hungarian Housing Intelligence

Software & Data Engineering Case Study | July 2024 – November 2026

Project Overview

What started in July 2024 as a Python web scraper tracking apartment prices in Debrecen has gradually evolved into a cloud-based data collection and analytics platform covering **47 locations across Hungary**.

The project combines approximately 2.5 years of historical data with automated data collection, database management, statistical analysis, and an interactive web dashboard. The longest historical series belongs to Debrecen; the 47-location expansion began later.

Technical Evolution

- **1. Data Collection & Processing (2024).** Developed a Python scraper using Requests and BeautifulSoup to extract property listing data, calculate price-per-square-meter values, and build a historical housing price index focused on 50–60 m² apartments.
- **2. Cloud Automation & Data Storage (2024–2025).** Integrated AWS Lambda and DynamoDB to automate data collection and enable long-term historical storage. Implemented MySQL integration for structured data management and time-series analysis.
- **3. Scaling to 47 Locations (2026).** Expanded the original Debrecen-focused system to 47 geographical locations, including all 23 Budapest districts. Redesigned the data pipeline with location-based indexing, UUIDv7 identifiers, batch database operations, and duplicate prevention.
- **4. Analytics & Desktop Application (2025–2026).** Built a Python desktop application using CustomTkinter, Pandas, and Matplotlib, featuring data exports, historical visualizations, and technical indicators such as moving averages, Bollinger Bands, RSI, and MACD.
- **5. Interactive Web Dashboard (November 2026).** Developed and published a Streamlit and Plotly dashboard, making the collected data accessible through an interactive web interface with location filtering, time-series analysis, comparative rankings, and downloadable datasets.

System Architecture

- **Data Collection:** Python · Requests · BeautifulSoup · AWS Lambda
- **Storage & Processing:** DynamoDB · MySQL · Pandas · SQL
- **Analytics & Visualization:** CustomTkinter · Matplotlib · Plotly · Streamlit · Power BI
- **Deployment & Version Control:** Git · GitHub · Streamlit Community Cloud

Key Engineering Challenges

- Maintaining a scraper despite changes in the source website’s HTML structure.
- Scaling an automated collection pipeline from one city to 47 locations.
- Integrating legacy historical data with a redesigned multi-location database.
- Ensuring data consistency, duplicate prevention, and synchronization between cloud and local databases.
- Separating data collection, storage, processing, and presentation into maintainable components.

Outcome & Future Development

The result is an end-to-end data engineering project demonstrating practical experience in **Python development, cloud automation, database design, ETL pipelines, data analytics, and interactive web application development**.

The platform is still evolving, with future plans for advanced market comparisons, improved statistical analysis, automated web data updates, and additional visualization features.

Project Links

- **Live Dashboard:** https://hungarian-housing-intelligence.streamlit.app
- **GitHub:** https://github.com/GregTakacsGergo/hungarian-housing-intelligence